

Indian Institute of Information Technology, Bhopal

Department of Computer Science & Engineering

Branch: CSE-(AI/CPS/CS/DS)

Semester: 3rd

Section: 3 & 4

Mid-Term Examination 08-09-2026 (Tuesday)

Maximum Marks: 20

Subject Code & Name:

CSE-(AI/CPS/CS/DS)-3002, Database

Management System

Time Allotted: 90 Minutes

- All questions are compulsory. Write your answers clearly and legibly. **There is no step marking. Only completely correct answers will be awarded full marks.**
- While solving, clearly mention the question numbers and their respective parts. Attempt the questions in the given sequence. All subparts of a question must be answered together in the same place.
- If any question appears ambiguous, do not worry; clearly state your assumptions in capital letters before answering.

1	A.	Consider two relations R1(A, B) with the tuples (1, 5), (3, 7) and R2(A, C) with the tuples (1, 7), (4, 9). Assume that a new type of join "full natural outer join" is performed between R1 and R2. This type of join is a combination of natural join and full outer join. Determine the resulting relation R(A, B, C). Show all the tuples in the resulting relation with step-by-step computation. Also explain natural join and full outer join briefly.	2
	B.	Consider two relations R(A,B) with the tuples (1, 3), (2, 6), (3, 8), (4, 10) and S(A, B) with the tuples (101, 3), (102, 6), (103, 8), (104, 10), (105, 12). Consider the following sql queries: query 1: select S.A from S where S.b in (select R.B from R where R.B > 5); query 2: select S.A from S where S.B = any (select R.B from R where R.B > 5); query 3: select S.A from S where S.B >= all (select R.B from R where R.B > 5); Assume that above queries are valid mysql queries. Determine the output of each query, show step-by-step computation, and identify which of the above queries are equivalent.	3
2	A✓	Suppose we have a relation declared by: create table R (name varchar(50) primary key, salary int check (salary <= 40000)); Assume that above query is a valid mysql query. Initially, the relation R contains the following tuples: R = {(tom, 10000), (joe, 20000), (sue, 30000)}. We try to execute the following sequence of modifications: (1) insert into R values ('fred', 12000); (2) update R set salary = 50000 where name = 'sue'; (3) insert into R values ('tom', 13000); (4) delete from R where name = 'joe'; at the end of these statements, what is the final table and sum of the salaries over all the tuples in R? Justify your answer	2

	B.	Consider three relations sailors(sid, sname, rating, age), boats(bid, bname, color), and reserves(sid, bid, day). The instances of the relations are as follows: sailors(sid, sname, rating, age) = {(22, Dustin, 7, 45.0), (29, Brutus, 1, 33.0), (31, Lubber, 8, 55.5), (32, Andy, 8, 25.5), (58, Rusty, 10, 35.0), (64, Horatio, 7, 35.0), (71, Zorba, 10, 16.0), (74, Horatio, 9, 35.0), (85, Art, 3, 25.5), (95, Bob, 3, 63.5)}. boats(bid, bname, color) = {(101, Interlake, blue), (102, Interlake, red), (103, Clipper, green), (104, Marine, red)}. reserves(sid, bid, day) = {(22, 101, 10/10/98), (22, 102, 10/10/98), (22, 103, 10/8/98), (22, 104, 10/7/98), (31, 102, 11/10/98), (31, 103, 11/6/98), (31, 104, 11/12/98), (64, 101, 9/5/98), (64, 102, 9/8/98), (74, 103, 9/8/98)}. Consider the following relational algebra query: $\pi_{sid}(\sigma_{age > 20}(\text{sailors})) - \pi_{sid}((\sigma_{color=red}(\text{boats})) \bowtie \text{reserves} \bowtie \text{sailors})$. Determine the number of rows returned by the above relational algebra query. Show a step by step approach for solving the problem.	3
3.	A.	Consider the following three table declarations for P(A), Q(B), and R(C), with referential integrity constraints: create table P (A int primary key); create table Q (B int primary key references P(A) on update cascade); create table R (C int primary key references Q(B) on update cascade); Assume that above commands are valid mysql commands. Initially, the tables contain the following tuples: P(A) = {(1), (2), (3), (4), (5), (6)} Q(B) = {(1), (2), (4), (6)} R(C) = {(1), (2), (6)} Suppose we execute the following modification command: UPDATE P SET A = A + 10 WHERE A < 5; What will be the output of the following query, assuming all referential integrity actions are performed? SELECT SUM(C) FROM R; Show your solution in step by step manner.	3
	B.	Consider the declarations of two relations R and S: create table S(C int primary key, D int); create table R(A int primary key, B int, check(B in (select C from S))); Assume that above commands are valid mysql commands. R(A, B) currently contains the four tuples (0, 4), (1, 5), (2, 4), and (3, 5). S(C, D) currently contains the four tuples (2, 10), (3, 11), (4, 12), and (5, 13). We perform the following operations. (a) Inserting (5, 2) into R. (b) Inserting (4, 6) into R. (c) Updating (0, 4) in R to be (0,0). (d) Inserting (1, 4) into R. Which operations will be successful or fail. Justify your answer for each operation.	2
4	A	A university offers many courses. Each student is uniquely identified by StudentID and has a StudentName. Each course is uniquely identified by CourseID and has a CourseName. A student can exist without enrolling in any course, but may enroll in many courses. A course can exist even if no student is enrolled, but may have many students. For every enrollment, the university stores the Semester and the Grade. Convert the above requirements into the corresponding relational schema. Clearly indicate the primary key(s), foreign key(s) and other constraints.	3
	B.	Let a Relation R have attributes {A, B, C, D, E} and "A" and "B" are the candidate keys. Then how many super keys are possible?	2